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$$f(x) = x^x$$

$$\Rightarrow y = x^x$$

$$\Rightarrow \ln y = \ln x^x = x \ln x$$

$$\Rightarrow (\ln y)' = (x \ln x)'$$

$$\Rightarrow \frac{y'}{y} = \ln x + x \cdot \frac{1}{x}$$

$$\Rightarrow y' = y (\ln x + 1)$$

$$\Rightarrow \boxed{f'(x) = x^x (\ln x + 1)}$$

References